MATERIAL SAFETY DATA SHEET

Version：V1.0

PU-608

1. Identification	
1.1 **Product name：**	Waterborne polyurethane dispersion PU-608
Chemical category：	Waterborne polyurethane resin dispersion
1.2 **Product use suggestions and restrictions：**	Used in leather finishing, textile coating, paper printing, adhesives and other industries.
1.3 **Supplier information:**	
Name of supplier：	Guangzhou Guanzhi New Material Technology Co., Ltd.
Supplier address:	Room 1106, Area A, Guangzhou International Business Incubator, No. 3 Juquan Road, Science City, Luogang District, Guangzhou
Tel：	86-20-82567990
Fax：	86-20-32214789

2.Hazards Identification	
2.1 **Emergency Overview**	Harmful if inhaled. Causes mild skin irritation. Causes eye irritation. May damage fertility or the unborn child. Causes damage to organs.
2.2 **GHS Hazard Classification：**	Reproductive toxicity (Category 1B), H360
2.3 **GHS Label Elements：**	F
GHS Pictograms：	[GHS07] [GHS08]
2.4 **Signal Word：**	Danger
2.5 **Hazard Statements：**	Hazard statement(s): H360: May damage the unborn child.
2.6 **Precautionary Statements：**	Precautionary statement(s): Prevention: P201: Obtain special instructions before use. P202: Do not handle until all safety precautions have been read and understood. P280: Wear protective gloves/protective clothing/eye protection/face protection.

2.Hazards Identification	
	Response: P308 + P313: IF exposed or concerned: Get medical advice/attention. Storage: P405: Store locked up. Disposal: P501: Dispose of contents/container in accordance with local/regional/national/international regulations.
2.7 Physical and Chemical Hazards：	Risk of cutaneous absorption of N-Methyl-2-pyrrolidone (NMP).

3.Composition/Information on Ingredients		
3.1 Product type：	Mixture	
3.2 Ingredients		
Chemical name	**CAS NO.**	**Concentration % (w/w)**
Polyurethane dispersion	9009-54-5	33.0~36.0
Deionized water	7732-18-5	50.0~54.0
N-Methyl-2-pyrrolidone (NMP)	872-50-4	13.0~14.0

4.First aid measures	
4.1 General measures：	Remove all contaminated clothing immediately.
4.2 Ingestion：	Do NOT induce vomiting. Seek medical attention immediately.
4.3 Eye contact：	Hold eyelids open and flush with plenty of warm water for at least 10 minutes. Consult an ophthalmologist.
4.4 Skin contact：	Wash immediately with plenty of soap and water. If skin reaction occurs, seek medical advice.
4.5 Inhalation：	If respiratory irritation occurs, seek medical advice.

5.Fire fighting measures	
5.1 Suitable extinguishing agents：	Carbon dioxide (CO2), foam, extinguishing powder. For large fires, use water spray.
5.2 Unsuitable extinguishing agents：	High-volume water jet.
5.3 Special hazards of substances or mixtures：	Combustion may release carbon monoxide, carbon dioxide, nitrogen oxides and trace amounts of hydrogen cyanide. Do not breathe fumes/smoke in case of fire or explosion.
5.4 Firefighting precautions protective equipment：	Firefighters must wear self-contained breathing apparatus. Do not allow contaminated fire-fighting water to enter soil, groundwater, or surface water.

6.Measures for accidental leakage	
6.1 Firefighting precautions protective equipment：	Wear protective equipment. Ensure adequate ventilation. Keep unauthorized persons away.
6.2 Environmental precautions：	Do not discharge into drains, wastewater, or soil.
6.3 Methods and material for containment and cleaning up：	Absorb with chemical absorbent material or dry sand if necessary, and store in tightly closed containers.

7.Operation and storage	
7.1 Precautions for safe handling：	Observe common chemical precautions during handling. Avoid contact with skin and eyes. Keep away from food, drink and tobacco. Wash hands before breaks and at the end of workday. Store work clothing separately. Change contaminated or soaked clothing.

7.Operation and storage	
7.2 Conditions for safe storage：	Keep containers tightly closed in a dry, well-ventilated place. To maintain product quality, strictly follow the storage conditions in our technical data sheet.

8.Exposure control / personal protection	
8.1 Exposure controls：	
Respiratory protection：	Respiratory protection equipment is required during spraying.
Hand protection：	Protective gloves are recommended.
Suitable material for protective gloves：	EN 374-3
Fluororubber - FKM:	Thickness ≥ 0.4 mm; Breakthrough time ≥ 480 min.
Butyl rubber - IIR：	Thickness ≥ 0.5 mm; Breakthrough time ≥ 480 min.
Nitrile rubber - NBR：	Thickness ≥ 0.35 mm; Breakthrough time ≥ 480 min.
Recommendation:	Contaminated gloves should be discarded.
Eye protection：	Wear protective goggles or face shield.
Body protection：	Wear appropriate protective clothing.
8.2 Engineering controls：	No exposure limit information is available under EC Directive 2006/121/EC.

9.Physical and chemical properties	
9.1 Appearance：	Transparent liquid
9.2 Odour threshold：	No data available.
9.3 pH value：	7-9
9.4 Ionicity：	Anionic
9.5 Initial boiling point：	Approx. 100℃
9.6 Flash point：	Not applicable
9.7 Evaporation rate：	No data available.
9.8 Flammability：	Not applicable
9.9 Heat of combustion：	Not applicable

9.Physical and chemical properties	
9.10 Saturated vapour pressure： pressure：	No data available.
9.11 Relative vapour density：	No data available.
9.12 Density：	1.0-1.1 g/cm^3
9.13 Solubility in water：	Miscible
9.14 Surface tension：	No data available.
9.15 log Pow (n-octanol/water partition coefficient)： partition coefficient)：	No data available.
9.16 Auto-ignition temperature： temperature：	Not applicable
9.17 Ignition temperature：	No data available.
9.18 Decomposition temperature： temperature：	No data available.
9.19 Dynamic viscosity：	< 250mPa.s
9.20 Explosive properties：	No data available.
9.21 Dust explosion class：	No data available.
9.22 Solid content：	35±2%
9.23 Other information：	The above data are not product specifications. Please refer to the Technical Data Sheet for specifications.

10.Stability and reactivity	
10.1 Chemical stability：	No decomposition if used according to specifications.
10.2 Hazardous decomposition products：	Flammable and harmful gases generated during thermal decomposition.
10.3 Possible hazardous reactions：	No hazardous reactions when stored and handled correctly.

11.Toxicity information		
No toxicological studies were available for this product.		
The following are toxicological data for N-Methyl-2-pyrrolidone (NMP):		
11.1 Acute toxicity：	**Oral:**	LD50 / Oral / Rat: 4,150mg/kg
	Inhalation:	LC50 / Inhalation / 1h / Rat: ＞ 5.1mg/l LC50 / Inhalation / 4h / Rat: ＞ 5.1mg/l
	Dermal:	LD50 / Oral / Rat: ＞5,000mg/kg
11.2 Primary skin irritation：	Species: Rabbit Classification: Irritating to skin Result: Causes skin irritation	
11.3 Primary eye irritation：	Species: Rabbit Classification: Irritating to eyes Result: Causes serious eye irritation	
11.4 Sensitization：	Species: Mouse Classification: Not a skin sensitizer Result: Did not cause sensitization in laboratory animals	
11.5 Mutagenicity：	No mutagenic effects observed in bacterial or mammalian cell cultures. No mutagenic effects observed in animal studies.	
11.6 Carcinogenicity：	No carcinogenic effects observed in animal studies.	
11.7 Reproductive toxicity：	**Fertility：**	Animal experiments showed risks of impaired fertility only at high doses.
11.8 Specific target organ toxicity (single/repeated exposure)：	May cause respiratory irritation.	
11.9 Aspiration hazard：	No data available.	
11.10 Additional information：	Risk of cutaneous absorption of	

11.Toxicity information	
	N-Methyl-2-pyrrolidone (NMP).

12.Ecological information	
No ecotoxicological studies are available for this product.	
The following are ecotoxicological data for N-Methyl-2-pyrrolidone (NMP):	
12.1 Ecotoxicity：	Toxicity to fish: LC50 / 96 h / Oncorhynchus mykiss (rainbow trout): 500mg/l Other: EC50 / 72 h / Algae: ＞500mg/l EC50 / 24 h / Daphnia magna: ＞1,000mg/l
12.2 Persistence and degradability：	Biodegradability: 73 %, 28 d, readily biodegradable.
12.3 Other adverse effects：	According to ecotoxicological data, the product is classified as non-hazardous to fish and daphnia.

13.Handling precautions	
Dispose of in accordance with applicable national standards and national or local regulations. Within the European Union, dispose of in accordance with the applicable provisions of the European Waste Catalogue (EWC).	
Disposal methods：	Empty containers as completely as possible (e.g., by pouring, scraping, or draining until “drip-dry”). They may be sent to an appropriate collection point under existing chemical industry recycling schemes. Recycle containers in accordance with national regulations and applicable environmental requirements. Do not discharge waste via wastewater.

14.Transportation information	
14.1 Road and rail transport：	Not classified as dangerous goods for land transport, in accordance with ADG regulations.
14.2 Maritime transport：	Not classified as dangerous goods for sea transport, in accordance with IMDG regulations.
14.3 Air transport：	Not classified as dangerous goods for air transport, in accordance with IATA regulations.
14.4 Special precautions for user：	Not regulated as dangerous goods. Temperature must not exceed +40℃ and must not fall below +5℃. Keep away from food, acids and bases.

15.Regulatory information
Relevant safety, health and environmental regulations for the substance or mixture
Other provisions：
Complies with the following regulations：
Regulations on the Safety Management of Hazardous Chemicals, State Council Decree No. 591
GB/T 16483 Safety data sheet for chemical products - Content and order of sections
GB 13690 General rules for classification and hazard communication of chemicals
GB 30000.2-29 Rules for classification and labelling of chemicals
GB 15258 General rules for preparation of precautionary label for chemicals

16.Other information
To the best of our knowledge, the information provided in this Safety Data Sheet is accurate as of the date of publication. The information given is designed only as guidance for safe handling, use, processing, storage, transportation, disposal, and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.